NAMMA SPACE · ROUND 1 · SUMMARY

All your options, and where we are

A one-page map of every way to build the 3D room — and the plan.

Start here

The goal: turn an ordinary phone video of a room into a **photorealistic, walk-through 3D web page** — no install, runs in a browser. **The core choice** is how to build the 3D model. There are 5 realistic ways; this pack has one simple page for each, plus our recommendation.

The 5 options at a glance



Higher = more photorealistic · Right = more effort/compute. The sweet spot (top-left) is cloud Gaussian splatting.

Option	Engine	Runs where	Look	Effort	Cost	Our result so far
1 ★ Cloud splatting Luma / Polycam cloud	3D Gaussian Splatting	Their cloud GPU	Photoreal	Low	Free tier	Not tried yet — recommended next
2 Phone splatting Scaniverse	3D Gaussian Splatting	On the phone	Photoreal	Low	Free	3 scans failed (untrained / empty)
3 Self-trained 3DGS COLMAP + gsplat/Brush	3D Gaussian Splatting	Colab / laptop GPU	Best possible	High	Free (Colab)	Not tried — power option
4 Photogrammetry / NeRF COLMAP + Instant-NGP	NeRF → textured mesh	Laptop/Colab GPU	Good	High	Free	Reviewed — outdated vs splats
5 monomap (ours) Depth Anything + Open3D	Mono-depth point fusion	Your laptop (offline)	Rough	Medium	Free	Working — cloud + mesh + floormap

What we have tried so far (the honest story)

Step we took	Result
Built **monomap**: phone video → mono-depth → point cloud + mesh + floorplan (fully offline)	Works, but not photorealistic
Realised Round 1 is scored on **photorealism + web roaming** → pivoted to **3D Gaussian Splatting**	Right strategy for scoring
Built a proper **web viewer app**: walk/orbit, loads & **saves** the scan, live progress bar	Ready & waiting for a scan
Scaniverse scan #1 → exported a **mesh** by mistake (no splat, no texture)	Wrong export type
Scaniverse scan #2 & #3 → splat format correct, but **empty** (training never converged)	On-device training failed
Decision: move capture to a **cloud** splat service (far more reliable) & keep monomap as the nav layer	Current plan

VERDICT FOR THE COMPETITION

Go with **Option 1 (cloud Gaussian splatting via Luma or Polycam)** for the Round-1 visual: photoreal, reliable, almost no compute on your side. Keep **Option 5 (monomap)** as the safety net and as the **navigation layer for Round 2** (it already outputs a floor-plan and occupancy map). Everything downstream — the SuperSplat clean-up and our web viewer — is **identical** no matter which option makes the splat.

Register by 15 Oct · Round 1 due 31 Oct · Capture: any tech allowed · Deliverables: PDF + web link + video + repo